THE SCIENCE OF HUNGER

Why we eat what we eat when we eat. Is it need, or plain greed?

While the purpose of hunger is to ensure that the body has enough energy to function and survive, it is possible to feel hunger without actually being hungry. Nutritionists term this second kind 'hedonic hunger', a state when one eats simply for the pleasure of eating. Also known as stress eating, food porn eating, or emotional eating, it makes one reach out for high-fat, high-sugar foods, particularly those that have previously made one 'feel good'—think burger, chocolate, a slice of cheesy pizza. The need to eat, or homeostatic hunger, which often makes itself manifest as weakness, a rumbling in the stomach, or even a headache if the hunger is not addressed in time, is an alert raised by the ghrelin hormone, which tells the brain it is time for food. In comparison, the high-calorie food typically binged on during a bout of hedonic hunger has been known to trigger dopamine, or pleasure centres in the brain. Such food is also known to reduce the brain's ability to regulate intake, making it difficult to stop at just one cookie, or cube of chocolate, or ice cream scoop. Semaglutide helps reduce hedonic hunger by working on hormones and promoting a feeling of fullness. Those on the medicine have also reported reduced craving for food, particularly that laden with sugar.



Illustration by **TANMOY CHAKRABORTY**



DR ISHI KHOSLA
Nutritionist

"THE MODERN GUT HAS ALREADY TAKEN SUCH A BEATING. TAKING MEDICINES THAT ALTER HOW YOUR DIGESTION WORKS COULD DO MORE HARM THAN GOOD. A DISTURBED GUT FLORA IS LINKED TO NEARLY 300 DIFFERENT MEDICAL CONDITIONS"

drugs' prohibitive cost could work against them. Novo Nordisk has not specified how it will price Ozempic and Wegovy if they get a licence in India. The company declined to comment on its plans to launch Ozempic in the country, though some reports in the media have hinted at a 2026 entry.

The solution could also lie in newer, better molecules of agonists. Besides Novo Nordisk and Eli Lilly, which are in a race to become the world's first trillion-dollar pharma companies, other drugmakers, too, are scrambling to corner a piece of the giant pie (see *The World of Agonists*). Lupin has been selling the dulaglutide product Aplevant in India under a collaboration with Eli Lilly. However, though a GLP-1 analog like semaglutide, it is not as effective in suppressing appetite. Sun Pharma is testing GL0034, a novel long-acting GLP-1 agonist, in non-obese and obese adults without diabetes. The CDSCO has approved Dr. Reddy's Labs to carry out a bioequivalence study of a semaglutide injection in India. Biocon, Alembic Pharma and Cipla, too, have expressed interest in entering this segment. Novo Nordisk has received approval from the Drug Controller General of India to conduct phase-III studies for its new weekly insulin injection, IcoSema, in India. IcoSema is a combination of insulin and semaglutide and is meant to control blood sugar levels in people with Type 2 diabetes.

The future certainly looks healthy for pharma firms, but the benefit to the diabetic and obese populations also promises to be immense. However, semaglutide needs to be used with caution and only if required, not just for the sake of vanity. ■